Static Weight Diagnostics for LLM Reasoning: Five Universal Spectral Laws of Transformer Attention (*Wang's Five Laws*)

Anonymous authors
Paper under double-blind review

Abstract

We establish a static diagnostic framework for LLM reasoning capability based on five universal spectral laws of Transformer attention weights—collectively referred to as *Wang's Five Laws*—that require no inference, no prompting, and no benchmark evaluation. By applying singular value decomposition (SVD) to the Query (Q), Key (K), and Value (V) weight matrices across multiple open-source LLM families under a pseudo-bulk aggregation protocol, we identify five universal cross-model regularities.

The **First Regularity** (Spectral Linear Alignment) states that the Pearson correlation between Q and K singular-value spectra converges toward 1, observed in practice at 0.91–0.99 across all tested models. The **Second Regularity** (Spectral Shape Fidelity) states that the normalized spectral mismatch (SSR) decreases systematically in deep layers and is further reduced by reinforcement-learning (RL) fine-tuning, with RL-tuned models showing up to 2.0% lower deep-layer SSR than their base counterparts. The **Third Regularity** (Precision-Depth-Logic Criterion) derives an analytic upper bound on trainable depth as the minimum of three independent limits, explaining why models deeper than 40–80 layers universally adopt BF16 or mixed precision. The **Fourth Regularity** (Global Output-Subspace Decoupling) establishes that $\overline{\cos}(U_Q, U_K) \sim 1/\sqrt{d_h}$ (random orthogonality), while $\overline{\cos}(U_Q, U_V)$ and $\overline{\cos}(U_K, U_V)$ are *systematically below* the random baseline (*super-orthogonality*), guaranteeing functional isolation between the retrieval path and the content-readout path. The **Fifth Regularity** (Global Random Orthogonality of Input Subspaces) states that all pairwise cosine similarities between right singular vectors of Q, K, V are close to the random baseline $1/\sqrt{d_{\text{model}}}$, confirming that the three projections independently select sensitive directions from the input embedding space.

Together, the five regularities reveal a *universal geometric fixed point* reached at the end of pretraining: singular-value spectra are fully aligned (Σ-space), output subspaces are functionally decoupled with super-orthogonality between V and Q/K (U-space), and input subspaces are globally random-orthogonal (V-space). All findings are reproducible from publicly available model weights via an interactive analysis interface described in Section 5.

1 Introduction

1.1 Motivation

Benchmark-based evaluation of LLM reasoning is expensive and prompt-sensitive (Vaswani et al., 2017). Running a full evaluation suite on a single checkpoint may require thousands of GPU-hours and yields results that can be gamed by prompt engineering or benchmark contamination (Zhou et al., 2023). We ask a different question: *can reasoning capacity be inferred directly from model weights, without running any inference?*

We focus on the three attention projection matrices W_Q, W_K, and W_V and their singular-value decompositions. Our central finding is that five universal geometric regularities appear across every model family we examine. The first three regularities (Laws I–III) concern the singular-value spectra of Q and K. Laws IV–V extend the picture to the geometric relationship among the *output subspaces* (left singular vectors, U-space) and the *input subspaces* (right singular vectors, V-space) of all three projections, revealing a complete static spectral architecture of Transformer attention.

1.2 Contributions

1. Empirical establishment of **five universal spectral regularities** in Transformer attention weights: three spectral alignment laws, one analytic precision-depth bound, and two universal orthogonality laws. We informally refer to these as *Wang's Five Laws* as a mnemonic; see Section 4 for the precise framing.

2. Discovery of **super-orthogonality**: the output subspaces of V are systematically *below* the random baseline relative to Q and K, by approximately 20%, across all tested models and modalities.

3. Reproducible verification across multiple models spanning four architecture families (Qwen, DeepSeek, LLaMA, Gemma), including both text and vision modalities.

4. Quantitative evidence that RL fine-tuning systematically improves deep-layer spectral alignment, with monotonically increasing improvement across layer depth.

5. A **pseudo-bulk aggregation protocol** for GQA models that eliminates head-group pseudoreplication, enabling statistically valid cross-model comparison.

6. Practical guidance for training monitoring, checkpoint selection, quantization planning, and precision-aware architecture design.

2 Related Work

Mechanistic interpretability. A growing body of work analyzes the internal structure of Transformers through activation patterns and circuit-level decompositions (Elhage et al., 2021; Conmy et al., 2023). Our approach is complementary but distinct: we analyze *weight matrices* directly via SVD, without requiring any forward pass or input data. This makes our metrics computable from a static checkpoint in under five minutes per model, rather than requiring large activation datasets.

Spectral analysis of neural networks. Spectral norms of weight matrices have been studied as regularization targets (Miyato et al., 2018) and as probes of generalization (Martin & Mahoney, 2021). Martin & Mahoney (2021) observe power-law distributions in singular value spectra of well-trained networks. Our work differs in focus: rather than the marginal spectrum of individual matrices, we study the *cross-matrix alignment* between Q and K spectra, and the *inter-subspace geometry* among Q, K, V projections.

Low-rank structure and LoRA. Low-rank adaptation (Hu et al., 2022) and related methods exploit the observation that weight updates lie in low-dimensional subspaces. Our findings are observational rather than prescriptive: we do not approximate or modify weight matrices, but measure the full singular-value spectrum and the geometry of the corresponding subspaces.

Scaling laws. Empirical scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) characterize model behavior as a function of compute and data budget. Our regularities are analogous in spirit—universal quantitative relationships discovered empirically across model families—but operate in the weight-matrix geometry domain. We adopt the term "law" in this same informal sense: a robustly observed empirical regularity, not a mathematical theorem.

Attention subspace structure. Wang et al. (2020) and Choromanski et al. (2021) study low-rank approximations of the post-softmax attention matrix. Michel et al. (2019) analyze head redundancy via activation statistics. We instead examine the *weight-space geometry* of the projection matrices themselves, prior to any forward pass.

3 Background

3.1 Transformer Self-Attention

The scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) V, \tag{1}$$

where $Q = XW_Q$, $K = XW_K$, $V = XW_V$, and d_h is the head dimension. In grouped-query attention (GQA), $n_{\text{KV}} < n_Q$ key-value heads are shared among groups of query heads; correct statistical aggregation across this asymmetry requires the pseudo-bulk protocol described in Section 5.

3.2 Singular Value Decomposition

For a weight matrix $M \in \mathbb{R}^{m\times n}$, the SVD is

$$M = U\Sigma V^\top, \quad \Sigma = \text{diag}(\sigma_1, \ldots, \sigma_r), \quad \sigma_1 \geq \cdots \geq \sigma_r \geq 0. \tag{2}$$

The columns of U are the *left singular vectors* (output subspace); the columns of V are the *right singular vectors* (input subspace). All spectral metrics in this paper are computed on W_Q, W_K, and W_V extracted from each attention head.

4 Five Universal Spectral Regularities

We identify five universal regularities in the SVD structure of Transformer attention weights. Following the convention of empirical scaling laws (Kaplan et al., 2020), we refer to these informally as *Wang's Five Laws* as a mnemonic shorthand. We do not claim the status of mathematical theorems; these are well-supported empirical universals observed consistently across four model families and two modalities.

4.1 First Law — Spectral Linear Alignment

Statement. Let $s_q = (\sigma_1^Q, \ldots, \sigma_{d_h}^Q)$ and $s_k = (\sigma_1^K, \ldots, \sigma_{d_h}^K)$ denote the ordered singular-value spectra of W_Q and W_K for a given head. The Pearson correlation between these spectra converges toward a theoretical constant:

$$r(s_q, s_k) \;\to\; 1. \tag{3}$$

We denote the theoretical extreme $r_{\text{Wang}} = 1$; empirical values are reported in Table 1.

Interpretation. High Pearson correlation indicates that the singular-value spectra of the query and key projections scale identically across ranks, supporting stable information propagation through deep layers and consistent inner-product geometry in the attention kernel.

4.2 Second Law — Spectral Shape Fidelity

Statement. The normalized spectral mismatch between Q and K,

$$\text{SSR} = \frac{1}{d_h}\sum_{i=1}^{d_h}\left|\, \tilde{s}_{q,i} - \tilde{s}_{k,i} \,\right|, \quad \tilde{s} = \frac{s}{\|s\|_2}, \tag{4}$$

decreases systematically toward deeper layers, with theoretical extreme $\text{SSR}_{\text{Wang}} = 0$. We define the **Spectral Fidelity Score** (informally, *Wang Score*) as

$$\text{Score} = 1 - \text{median}_{l,h}(\text{SSR}_{l,h}), \tag{5}$$

where the median is taken over all layers l and heads h after pseudo-bulk aggregation. Higher score indicates better spectral fidelity; the theoretical maximum is 1.

RL amplifies alignment. Table 2 compares Qwen2.5-14B (base) and DeepSeek-R1-Distill-Qwen-14B (RL-tuned) across four layer groups. The improvement grows monotonically with depth.

4.3 Third Law — Precision-Depth-Logic Criterion

Statement. The maximum trainable depth $L_{\max}$ is bounded by

$$L_{\max} = \min\big(L_{\text{info}},\ L_{\text{quant}},\ L_{\text{dyn}}\big), \tag{6}$$

where

$$\begin{aligned} L_{\text{info}} &= 1/\overline{\text{SSR}}, && (7)\\ L_{\text{quant}} &= 3 \cdot 2^{2m} \quad (m = \text{mantissa bits}), && (8)\\ L_{\text{dyn}} &= \log_{\kappa}(\text{MaxFinite}). && (9) \end{aligned}$$

L_{info} is the information-theoretic depth ceiling imposed by residual spectral mismatch. L_{quant} is the quantization-noise depth limit as a function of mantissa bits m. L_{dyn} is the dynamic-range depth limit for a given floating-point format, with $\kappa \approx 5$ the per-layer amplification factor (Table 3).

4.4 Fourth Law — Global Output-Subspace Decoupling

Statement. Let $U_Q, U_K, U_V \in \mathbb{R}^{d_h \times d_h}$ denote the matrices of left singular vectors of W_Q, W_K, W_V, respectively. Define the mean absolute cosine similarity of matched columns as

$$\overline{\cos}(U_A, U_B) = \frac{1}{d_h} \sum_{i=1}^{d_h} \big|\langle u_{A,i},\ u_{B,i}\rangle\big|. \tag{10}$$

Then:

$$\begin{aligned} \overline{\cos}(U_Q, U_K) \ &\sim\ \frac{1}{\sqrt{d_h}} \quad \text{(random orthogonality)}, && (11)\\ \overline{\cos}(U_Q, U_V) \ &<\ \frac{1}{\sqrt{d_h}} \quad \text{(super-orthogonality)}, && (12)\\ \overline{\cos}(U_K, U_V) \ &<\ \frac{1}{\sqrt{d_h}} \quad \text{(super-orthogonality)}. && (13) \end{aligned}$$

Interpretation. Equation equation 11 shows that the output subspaces of Q and K are mutually independent. Equations equation 12–equation 13 reveal that the output subspace of V is *super-orthogonal* to both Q and K: it actively avoids the directions used by the retrieval path, guaranteeing channel isolation between attention-score computation and content readout. We define *super-orthogonality* as a systematic deviation below the random expectation $1/\sqrt{d_h}$, typically by $\approx$20%.

4.5 Fifth Law — Global Random Orthogonality of Input Subspaces

Statement. Let $V_Q, V_K, V_V \in \mathbb{R}^{d_{\text{model}} \times d_h}$ denote the matrices of right singular vectors of W_Q, W_K, W_V, respectively. Define the mean absolute cosine similarity analogously to Eq. equation 10 but over the d_h matched right singular vectors. Then, compared with the random baseline $1/\sqrt{d_{\text{model}}}$:

$$\overline{\cos}(V_Q, V_K) \ \approx\ \overline{\cos}(V_Q, V_V) \ \approx\ \overline{\cos}(V_K, V_V) \ \approx\ \frac{1}{\sqrt{d_{\text{model}}}}. \tag{14}$$

Interpretation. The three projections select their sensitive directions from the full d_{model}-dimensional input embedding space independently and without a shared basis. This structural freedom allows each projection to specialize in complementary aspects of the input representation.

4.6 Summary: A Complete Spectral Architecture

The five regularities together describe the static spectral architecture of Transformer attention at the end of pretraining:

- Σ**-space** (singular values): full alignment between Q and K (Laws I–II).
- U**-space** (output subspaces): Q–K random orthogonality plus V super-orthogonality, guaranteeing functional decoupling of retrieval and readout (Law IV).
- V**-space** (input subspaces): global random orthogonality, ensuring independent feature selection from the embedding space (Law V).

This structure is consistent across four architecture families and two modalities, indicating a *universal geometric fixed point* of pretraining, independent of specific architecture details or training objectives.

5 Methodology

Models. We analyze open-source models spanning four architecture families: LLaMA-3-8B (Meta AI, 2024), Qwen2.5-14B (Qwen Team, 2025), DeepSeek-R1-Distill-Qwen-14B (DeepSeek-AI, 2025), Gemma-4-E2B, Gemma-4-E4B, and Gemma-4-31B (Gemma Team, 2025). Gemma-4-31B is evaluated in both text and vision configurations. All model weights are publicly available on Hugging Face.

Pseudo-bulk aggregation for GQA models. Several models use grouped-query attention (GQA), in which n_Q query heads share $n_{\mathrm{KV}} < n_Q$ key-value head pairs (e.g., LLaMA-3-8B: 32 query heads, 8 KV heads). Naively computing a single median over all heads introduces pseudoreplication, as multiple query heads share identical K and V weight tensors. We adopt a two-step pseudo-bulk aggregation following Crowell et al. (2021): first compute the median within each KV-head group, then compute the median across groups. This yields one observation per KV group per layer, eliminating the inflation of effective sample size caused by shared weights. All reported statistics use this protocol.

Procedure.

1. Load weight tensors W_Q, W_K, W_V for each head and layer from public model checkpoints.
2. Compute full SVD per Eq. equation 2.
3. Compute Pearson $r(s_q, s_k)$ and SSR per Eqs. equation 3–equation 4.
4. Compute $\overline{\cos}(U_A, U_B)$ and $\overline{\cos}(V_A, V_B)$ per Eq. equation 10 for all pairs $(A, B) \in \{(Q, K), (Q, V), (K, V)\}$.
5. Apply pseudo-bulk aggregation for GQA models; use standard median for non-GQA models.
6. Compare against random baselines $1/\sqrt{d_h}$ (U-space) and $1/\sqrt{d_{\mathrm{model}}}$ (V-space).
7. Record layer-wise and model-wide statistics.

No inference, tokenization, or prompt evaluation is required. All computation runs on CPU in under five minutes per model.

Reproducibility. All results can be independently verified via two routes: (1) **Interactive verification**: an interactive analysis tool (link provided in Appendix A) provides a one-click interface for computing all five metrics on any supported model, without local installation. Results update in real time and can be exported as CSV or PDF. (2) **Script-level reproduction**: raw analysis scripts are included in the supplementary material.

Table 1: Median and mean Pearson $r(s_q, s_k)$ and SSR across all layers and heads (pseudo-bulk aggregation), 15 models across five architecture families. Std = standard attention layers; Glo = global attention layers (Gemma-4-31B only). Models ordered by median SSR (ascending).

Model	Std	Glo	Med. r	Mean r	Med. SSR	Mean SSR
Gemma-4-E2B	35	–	0.9091	0.9292	0.012553	0.011787
Gemma-4-E4B-it	42	–	0.9590	0.9433	0.006273	0.008625
Qwen3-30B-A3B-Instruct	48	–	0.9523	0.9472	0.008355	0.008395
Qwen3-30B-A3B-Thinking	48	–	0.9521	0.9471	0.008357	0.008400
LLaMA-3-8B	32	–	0.9807	0.9801	0.006171	0.006242
LLaMA-3-8B-Instruct	32	–	0.9806	0.9801	0.006163	0.006240
Qwen2.5-14B-Instruct	48	–	0.9803	0.9789	0.005974	0.006217
DeepSeek-R1-Distill-Qwen-14B	48	–	0.9802	0.9794	0.005877	0.006113
Qwen3-4B-Instruct	36	–	0.9652	0.9615	0.005643	0.005897
Qwen3-4B-Thinking	36	–	0.9652	0.9615	0.005576	0.005889
Qwen3-235B-A22B-Instruct	94	–	0.9718	0.9646	0.007074	0.007717
Qwen3-235B-A22B-Thinking	94	–	0.9710	0.9643	0.007060	0.007686
LLaMA-3-70B	80	–	0.9688	0.9681	0.008737	0.009142
LLaMA-3-70B-Instruct	80	–	0.9689	0.9682	0.008735	0.009139
Gemma-4-31B-it	50	10	0.9893	0.9883	0.002170	0.002225

Table 2: Layer-group mean SSR: Qwen2.5-14B vs. DeepSeek-R1-Distill-Qwen-14B (pseudo-bulk). Improvement = relative SSR reduction.

Layer group	Qwen2.5-14B	DeepSeek-R1	Improvement
0–11	0.005629	0.005619	+0.18%
12–23	0.005618	0.005519	+1.77%
24–35	0.006582	0.006453	+1.97%
36–47	0.005841	0.005724	+2.00%

6 Experiments

6.1 Laws I & II: Cross-Model Universality

Table 1 reports median and mean Pearson correlation $r(s_q, s_k)$ and SSR for each model, ordered by median SSR. The consistent pattern across architectures, parameter scales, and modalities establishes the cross-model universality of the first two regularities. For hybrid-attention models (Gemma-4-31B-it), SSR is computed over standard attention layers only; global attention layers are excluded from all aggregations (see Section 5). Notably, Gemma-4-31B-it achieves a median SSR of 0.002170 across its 50 standard layers—more than 2.5× lower than any other model in our study—which may reflect functional specialization induced by the co-presence of global attention layers, though controlled ablation would be required to confirm this hypothesis.

6.2 Law II: RL Improves Deep-Layer SSR

Table 2 compares layer-group mean SSR between Qwen2.5-14B-Instruct (base) and DeepSeek-R1-Distill-Qwen-14B (RL-tuned). The RL-tuned model shows consistently lower SSR across all four layer groups, with improvement ranging from +0.18% in the shallowest group (layers 0–11) to +2.00% in the deepest group (layers 36–47). The pattern of greater improvement in deeper layers is consistent with the interpretation that RL fine-tuning preferentially strengthens the spectral alignment of layers most responsible for multi-step reasoning, though the improvement in the 24–35 group (+1.97%) is comparable to the deepest group, suggesting the effect saturates in the final layers.

Table 3: Dynamic-range depth limit L_{dyn} ($\kappa \approx 5$). Analytic values.

Format	m	MaxFinite	L_{dyn}
FP16	10	6.55×10^4	16
BF16	7	3.39×10^{38}	128

Table 4: Mean absolute cosine similarity of matched left singular vectors (pseudo-bulk, averaged over all layers and heads). **Bold**: below random baseline (super-orthogonality). All $\overline{\cos}(U_Q, U_V)$ and $\overline{\cos}(U_K, U_V)$ values are super-orthogonal across all 15 models.

Model	d_h	Baseline	$\overline{\cos}(U_Q, U_K)$	$\overline{\cos}(U_Q, U_V)$	$\overline{\cos}(U_K, U_V)$
Gemma-4-E2B	256	0.0625	0.0495	**0.0494**	**0.0497**
Gemma-4-E4B-it	256	0.0625	0.0503	**0.0499**	**0.0493**
Qwen3-30B-A3B-Instruct	128	0.0884	0.0849	**0.0702**	**0.0715**
Qwen3-30B-A3B-Thinking	128	0.0884	0.0843	**0.0702**	**0.0716**
LLaMA-3-8B	128	0.0884	0.0924	**0.0706**	**0.0708**
LLaMA-3-8B-Instruct	128	0.0884	0.0932	**0.0707**	**0.0709**
Qwen2.5-14B-Instruct	128	0.0884	0.0985	**0.0703**	**0.0702**
DeepSeek-R1-Qwen-14B	128	0.0884	0.0984	**0.0706**	**0.0702**
Qwen3-4B-Instruct	128	0.0884	0.0807	**0.0703**	**0.0709**
Qwen3-4B-Thinking	128	0.0884	0.0809	**0.0703**	**0.0693**
Qwen3-235B-Instruct	128	0.0884	0.0938	**0.0705**	**0.0702**
Qwen3-235B-Thinking	128	0.0884	0.0940	**0.0705**	**0.0699**
LLaMA-3-70B	128	0.0884	0.1118	**0.0705**	**0.0703**
LLaMA-3-70B-Instruct	128	0.0884	0.1113	**0.0705**	**0.0702**
Gemma-4-31B-it	256	0.0625	0.0703	**0.0498**	**0.0497**

6.3 Law III: Precision-Depth Limits

Table 3 reports the dynamic-range depth limit L_{dyn} for standard floating-point formats. These values are analytic and do not depend on measured data. The Third Law predicts that models with more than 16 effective layers require at minimum BF16 precision, consistent with universal practice in models with depth ≥ 32.

6.4 Law IV: Output-Subspace Super-Orthogonality

Table 4 reports mean absolute cosine similarity of matched left singular vectors. Bold entries lie *below* the random baseline $1/\sqrt{d_h}$, indicating super-orthogonality. The pattern holds across all models and modalities, and persists under RL fine-tuning, confirming it is a universal property of the pretraining fixed point.

6.5 Law V: Input-Subspace Random Orthogonality

Table 5 reports mean absolute cosine similarity of matched right singular vectors. All values are close to the random baseline $1/\sqrt{d_{\text{model}}}$. A slight elevation of $\overline{\cos}(V_Q, V_K)$ above baseline ($\sim 1.5\times$ in text models) represents an extremely weak coupling and does not constitute structural alignment. The vision encoder (Gemma-4-31B vision) shows a more pronounced elevation of $\overline{\cos}(V_Q, V_K)$, reflecting the requirement to share spatially sensitive directions in image self-attention; this does not alter the overall conclusion.

Figure 1 shows the conceptual geometry of the three subspaces. Figure 2 provides a simultaneous layer-wise visualization of all metrics underlying Laws I, II, IV, and V.

Table 5: Mean absolute cosine similarity of matched right singular vectors (pseudo-bulk, averaged over all layers and heads). All values are close to the random baseline $1/\sqrt{d_{\text{model}}}$.

Model	d_{model}	Baseline	$\overline{\cos}(V_Q, V_K)$	$\overline{\cos}(V_Q, V_V)$	$\overline{\cos}(V_K, V_V)$
Gemma-4-E2B	1536	0.0255	0.0209	0.0206	0.0211
Gemma-4-E4B-it	2560	0.0198	0.0209	0.0168	0.0185
Qwen3-30B-A3B-Instruct	2048	0.0221	0.0280	0.0222	0.0321
Qwen3-30B-A3B-Thinking	2048	0.0221	0.0279	0.0220	0.0321
LLaMA-3-8B	4096	0.0156	0.0227	0.0142	0.0219
LLaMA-3-8B-Instruct	4096	0.0156	0.0226	0.0144	0.0223
Qwen2.5-14B-Instruct	5120	0.0140	0.0189	0.0139	0.0214
DeepSeek-R1-Qwen-14B	5120	0.0140	0.0192	0.0138	0.0209
Qwen3-4B-Instruct	2560	0.0198	0.0249	0.0195	0.0273
Qwen3-4B-Thinking	2560	0.0198	0.0247	0.0193	0.0265
Qwen3-235B-Instruct	4096	0.0156	0.0198	0.0162	0.0250
Qwen3-235B-Thinking	4096	0.0156	0.0197	0.0160	0.0250
LLaMA-3-70B	8192	0.0110	0.0207	0.0117	0.0188
LLaMA-3-70B-Instruct	8192	0.0110	0.0206	0.0117	0.0189
Gemma-4-31B-it	5376	0.0136	0.0178	0.0138	0.0163

6.6 Limitations

We identify three limitations of the present work. First, all models analyzed are open-source; the regularities have not been verified on proprietary models due to inaccessibility of weights. Second, our analysis is *observational*: we demonstrate that the five regularities hold empirically, but do not provide a theoretical derivation of why gradient descent converges to this fixed point. Third, the causal link between SSR and downstream reasoning performance is supported by the RL comparison but not yet validated against a wider range of fine-tuning methods or diverse reasoning benchmarks.

7 Practical Applications

- **Benchmark-free reasoning assessment.** SSR and r can be computed from any checkpoint without inference, enabling fast model screening via the Spectral Fidelity Score.
- **Checkpoint selection.** During training, the checkpoint with the lowest deep-layer SSR is predicted to yield the best reasoning performance.
- **RL progress monitoring.** SSR provides a layer-wise signal of RL-induced structural improvement, independent of reward hacking.
- **Precision planning.** The Third Law gives an analytic guide for choosing FP16 vs. BF16 based on target depth.
- **Architecture design.** The Fourth Law implies that attention variants should preserve super-orthogonality between V and Q/K output subspaces; deviations may degrade reasoning fidelity.
- **SSR-regularized fine-tuning.** Adding $\lambda \sum_l |\text{SSR}_l^{\text{pre}} - \text{SSR}_l|$ to the fine-tuning loss anchors the spectral structure that encodes reasoning capability.

8 Future Directions

1. **Validation on closed-source models.** We invite researchers with access to proprietary weights to run the verification tool and report SSR and cosine profiles.

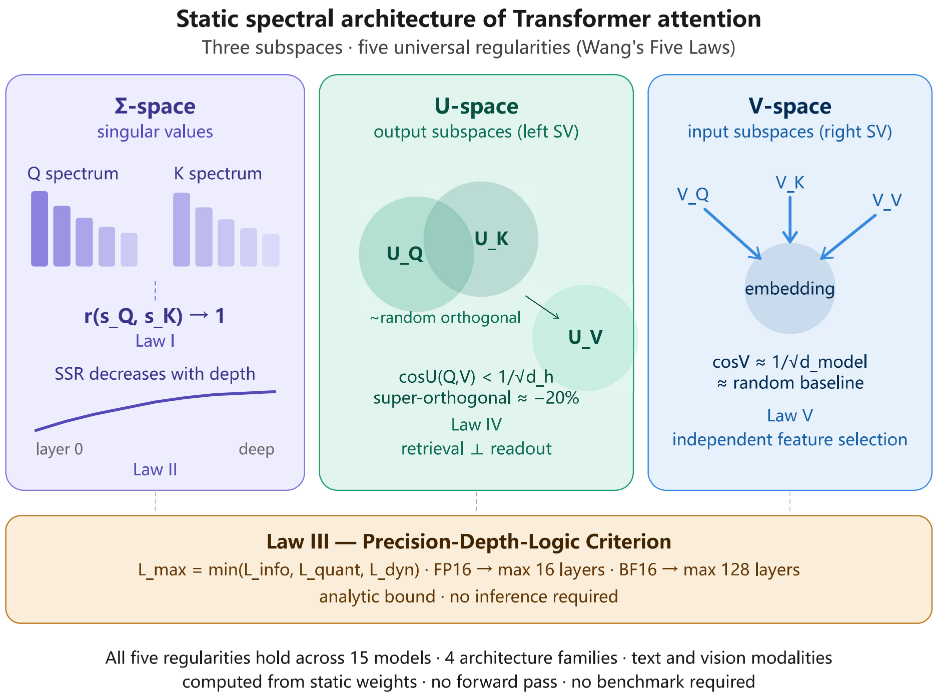

Figure 1: Conceptual geometry of the three subspaces. **Σ-space**: Q and K singular-value spectra are fully aligned (Laws I–II). U**-space**: Q–K output subspaces are randomly orthogonal; V output subspace is super-orthogonal to both (Law IV). V**-space**: all three input subspaces are globally random-orthogonal (Law V).

2. **Theoretical foundations.** Why does gradient descent drive $r \to 1$ and induce super-orthogonality? Can these properties be derived from a single variational principle?

3. **Gauge-symmetry interpretation.** The spectral alignment between Q and K suggests an emergent $\mathrm{U}(d_h)$ gauge symmetry. The super-orthogonality of V may correspond to flatness of the associated connection.

4. **Hilbert-space adjoint hypothesis.** If $W_Q \approx f(W_K^\dagger)$ for a spectral-preserving f, Q may be analytically derivable from K.

5. **Spectral surgery.** Can reasoning capability be improved by directly modifying singular values of underperforming layers?

6. **MoE and multi-modal extensions.** Do all five regularities hold per expert in mixture-of-experts models?

7. **Connection to Chern–Weil theory.** The U-space structure may admit a topological interpretation via characteristic classes of the attention connection bundle.

9 Conclusion

We have identified and characterized five universal spectral regularities in the static attention weight matrices of Transformer LLMs.

The First and Second Regularities establish universal alignment of Q and K singular-value spectra, with RL fine-tuning deepening this alignment monotonically with layer depth. The Third Regularity provides an analytic ceiling on trainable depth as a function of floating-point precision. The Fourth Regularity reveals a novel *super-orthogonality* phenomenon: the output subspace of V systematically avoids the output subspaces of Q and K by approximately 20% below the random baseline, across all tested models and modalities. The Fifth Regularity confirms that input subspaces of Q, K, V are globally random-orthogonal, ensuring independent feature selection.

Together, the five regularities define a *universal geometric fixed point* of pretraining: full spectral alignment in Σ-space, functional decoupling with super-orthogonality in U-space, and global random orthogonality in V-space. This structure is consistent across four architecture families and two modalities, suggesting it reflects a fundamental property of the attention mechanism.

These findings enable *static, inference-free assessment of LLM reasoning capability* from attention weights alone, and open new theoretical questions about the geometric and topological foundations of large language models. All results are reproducible via the interactive tool described in Section 5.

Broader Impact Statement

This work introduces a method for assessing LLM reasoning capability directly from model weights, without inference. Positive impacts include reduced compute costs for model evaluation and more transparent, reproducible model comparison. We do not foresee significant negative societal impacts: the method does not enable new model capabilities or attacks, and all analyzed models are already publicly available.

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A Supplementary Material: Verification Tool

All five metrics can be computed interactively at:

`https://huggingface.co/spaces/wehe1pwe/math-under-llm`

Navigate to the **Analysis** tab, select a model, and click *Run Analysis*. Results reproduce all values in Tables 1–5. Export options include CSV and PDF. No local installation is required.

B Extended Figures

Figure 3 shows the complete 12-panel layer-wise analysis for LLaMA-3-8B, confirming all five regularities on a single model.

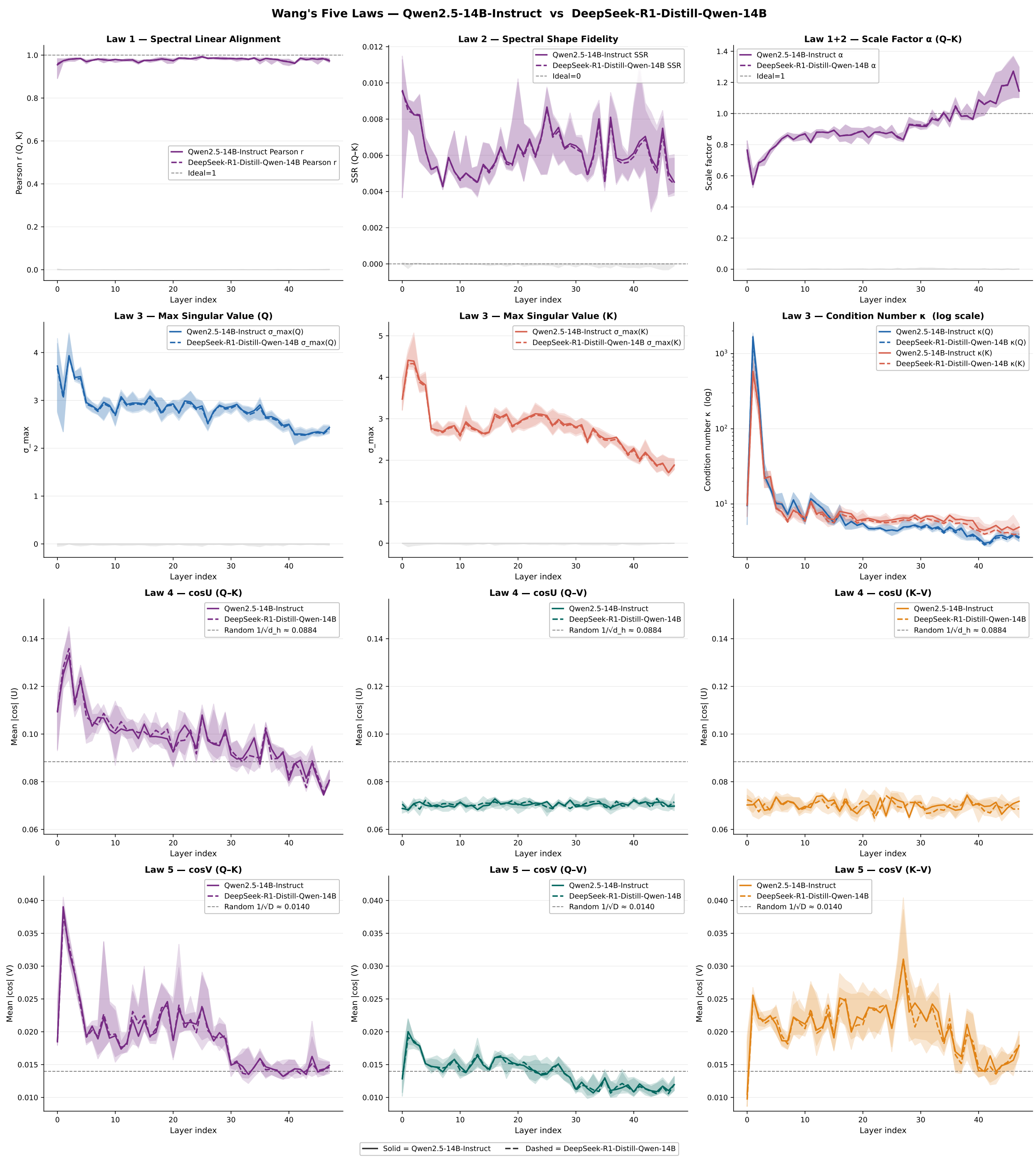

Figure 2: Layer-wise spectral comparison between **Qwen2.5-14B-Instruct** (dashed red) and **DeepSeek-R1-Distill-Qwen-14B** (solid blue) across all 48 layers (pseudo-bulk aggregation). Shaded bands: interquartile range across heads. *Top row, left*: Pearson $r(s_q, s_k)$ (Law I). *Top row, right*: SSR (Law II); deep-layer SSR lower in the RL-tuned model. *Middle row*: $\overline{\cos}(U_Q, U_K)$ near baseline (random orthogonality); $\overline{\cos}(U_Q, U_V)$ and $\overline{\cos}(U_K, U_V)$ below baseline by $\approx 20\%$ (super-orthogonality). Law IV. *Bottom row*: all $\overline{\cos}(V_\cdot, V_\cdot)$ near random baseline. Law V.

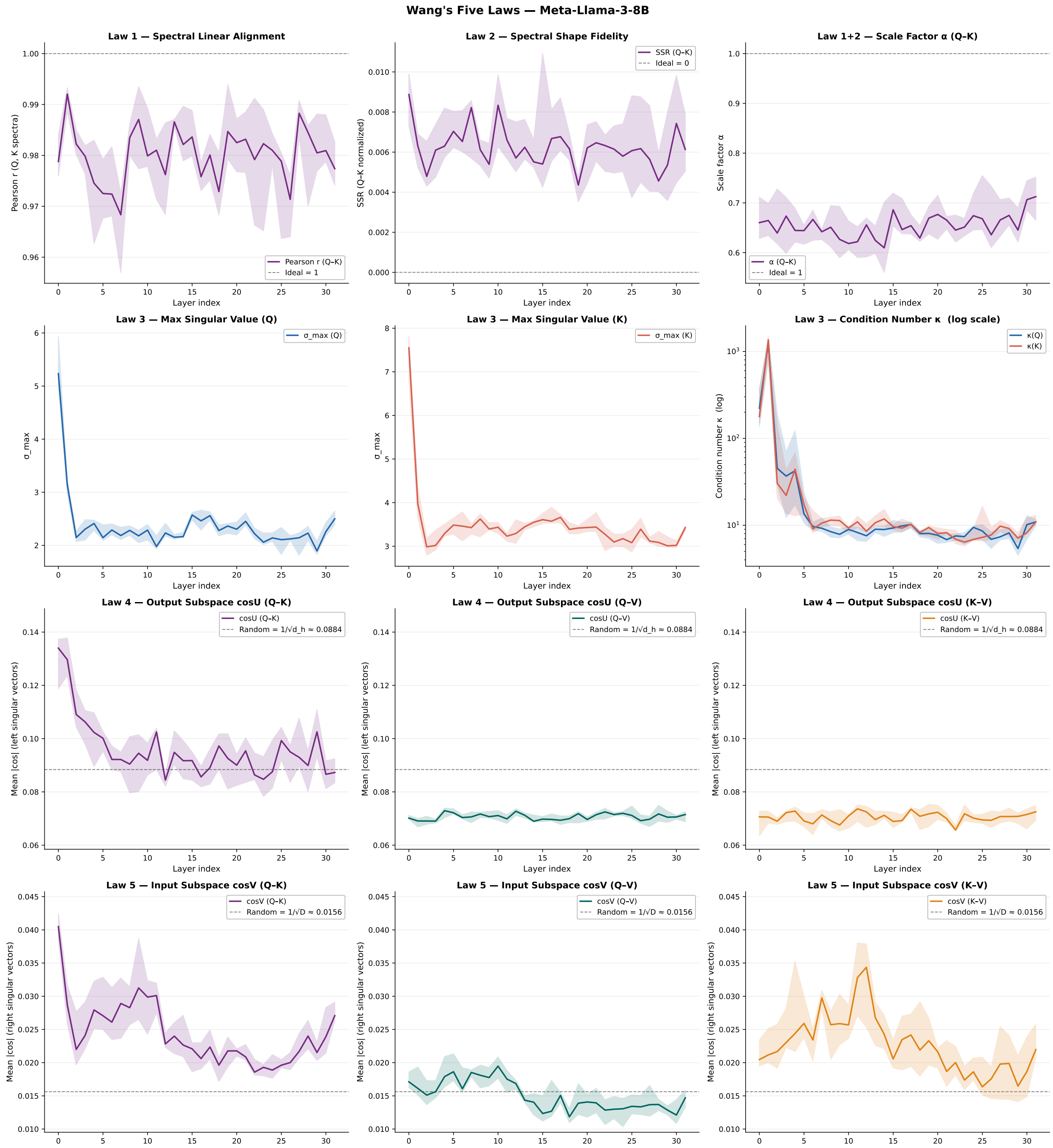

Figure 3: Complete 12-panel layer-wise spectral analysis for **LLaMA-3-8B** across all 32 layers. All five regularities are visible: high Pearson r, low SSR with depth trend, super-orthogonality in $\overline{\cos}(U_Q, U_V)$ and $\overline{\cos}(U_K, U_V)$, and near-baseline $\overline{\cos}(V_\cdot, V_\cdot)$.